

Equation (2.5) are farther to the left of the j axis. The first three terms are termed *nondominant*, whereas the fourth term

$$A_2 e^{-t} \cos(6t + \theta_2) \quad [2.7]$$

is termed *dominant*. This fourth pair of dominant roots determines the behavior of the system more than the nondominant roots. This is because the first three terms in Equation (2.6) will decay to zero in such a short period of time that the time response will be determined largely by the fourth term, shown in Equation (2.7). The system behavior can often be approximated by the dominant roots, or the "least-damped roots" ($\zeta_2 = 0.16$). This concept is important in higher-order systems, which are usually dominated by the pair of "least-damped roots" located closest to the j axis of the s plane. For roots to be dominant, the real parts of the complex conjugate roots should be approximately 20% of the real part of any other roots of the system. The relative magnitude of the residues (constant multiplier) of each root must also be considered. These residues, of course, depend upon the location of the zeros of the system function. The concept of dominant roots is useful to estimate the system response, but it should be used with care.

□ Computer-Aided Learning

C2.7 Use MATLAB "ezplot" command to get the zero-state impulse response for the two transfer functions given in Section 2.4.

```
Ans. (a) syms t s
        Ya=(-8*s^2+5)/(s+1)/(s+4)/(s^2+4*s+13)
        ya=ilaplace(Ya)
        ezplot(ya,[0,4])

        (b) syms t s
        Yb=(-7*s^3+6*s^2+9*s+23)/(s+9)/(s+12)
            /(s^2+12*s+40)/(s^2+2*s+37)
        yb=ilaplace(Yb)
        ezplot(yb,[0,5])
```

2.5 Stability Testing

2.5.1 Coefficient Tests

For first- and second-order systems, stability (see Section 1.12) is determined by inspection of the characteristic polynomial. (A first- or second-order polynomial has all roots in the left half of the complex plane if and only if all polynomial coefficients have the same algebraic sign. For example,)

$$3s^2 + s + 10$$

is the characteristic polynomial of a stable system, while

$$3s^2 + s - 10$$

represents an unstable system.

First- and Second-order stable systems.

For higher-order polynomials, representing higher-order systems, the algebraic signs of the polynomial coefficients may or may not yield information as to stability. A polynomial with all roots in the left half-plane (LHP) has factors of the form

$$(s + a) \quad a > 0 \quad (\text{real axis root in the LHP})$$

and

$$(s^2 + bs + c) \quad b > 0 \quad \text{and} \quad c > 0 \quad (\text{two LHP roots, perhaps complex conjugate})$$

When multiplied out, such a polynomial must have all coefficients of the same algebraic sign, all positive or all negative. No coefficient can be zero ("missing") in a system with LHP roots because there are no minus signs involved and thus no way for a coefficient to be canceled.

If imaginary axis roots exist in the polynomial, factors of the following forms can be present, in addition to the others:

$$(s) \quad (\text{root at the origin})$$

and

$$(s^2 + a) \quad a > 0 \quad (\text{complex conjugate roots on the imaginary axis})$$

With such roots present, all polynomial coefficients must be of the same algebraic sign, but some coefficients can be zero.

Right half-plane (RHP) roots involve factors of the form

$$(s - a) \quad a > 0 \quad (\text{real axis root in the RHP})$$

and

$$(s^2 - as + b) \quad a > 0 \quad \text{and} \quad b > 0 \quad (\text{two roots in the RHP, perhaps complex conjugate})$$

The presence of such factors may or may not cause differing algebraic signs of the coefficients and (by cancellation) zero coefficients.

Table 2.1 summarizes the information conveyed by these coefficient tests. For example, the polynomial

$$7s^6 + 5s^4 - 3s^3 - 2s^2 + s + 10$$

Negative signs indicate RHP roots (unstable system).

Table 2.1 Polynomial Coefficient Tests

Properties of the Polynomial Coefficients	Conclusion About Roots from the Coefficient Test
Differing algebraic signs	At least one RHP root
Zero-valued coefficients	Imaginary axis or RHP roots or both
All of the same algebraic sign, none zero	No direct information

has one or more RHP roots, indicated by the differing algebraic signs of the coefficients. Examination of the coefficient signs yields no information about root locations for the following polynomial:

$$8s^5 + 6s^4 + 3s^3 + 2s^2 + 7s + 10$$

The polynomial

$$s^6 + 3s^5 + 2s^4 + 8s^2 + 3s + 17$$

has imaginary axis roots or RHP roots or both, indicated by the missing s^3 term. Imaginary axis roots in the polynomial, if they are present, are complex conjugate, since if there were an imaginary axis root at $s = 0$, s would be a factor of the polynomial.

Implication of missing terms.

2.5.2 Routh-Hurwitz Testing

The Routh-Hurwitz test is a numerical procedure for determining the numbers of right half-plane (RHP) and imaginary axis (IA) roots of polynomial. The characteristic polynomial, which is the denominator of $T(s)$ is expressed as follows:

$$p(s) = a_n s^n + a_{n-1} s^{n-1} + \cdots + a_1 s + a_0 \quad [2.8]$$

An array of numbers a_i is established as follows:

$$\begin{array}{l} s^n \\ s^{n-1} \\ s^{n-2} \\ s^{n-3} \\ s^{n-4} \\ \vdots \\ s^0 \end{array} \left| \begin{array}{ccc} a_n & a_{n-2} & a_{n-4} \\ a_{n-1} & a_{n-3} & a_{n-5} \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \quad \dots \\ d_1 & d_2 & d_3 \end{array} \right.$$

[2.9] *Routh array format.*

The first row starts with the highest-order constant a_n , and then every other constant is set forth. The second row starts with the second-highest-order constant a_{n-1} and then every other constant. The first two rows are formed from the coefficients of $p(s)$. The subsequent rows are formed as follows:

$$b_1 = \frac{- \begin{vmatrix} a_n & a_{n-2} \\ a_{n-1} & a_{n-3} \end{vmatrix}}{a_{n-1}} = \frac{a_{n-1}a_{n-2} - a_n a_{n-3}}{a_{n-1}} \quad [2.10]$$

$$b_2 = \frac{- \begin{vmatrix} a_n & a_{n-4} \\ a_{n-1} & a_{n-5} \end{vmatrix}}{a_{n-1}} = \frac{a_{n-1}a_{n-4} - a_n a_{n-5}}{a_{n-1}} \quad [2.11]$$

And the fourth row is formed as follows:

$$c_1 = \frac{- \begin{vmatrix} a_{n-1} & a_{n-3} \\ b_1 & b_2 \end{vmatrix}}{b_1} = \frac{b_1 a_{n-3} - b_2 a_{n-1}}{b_1} \quad [2.12]$$

$$c_2 = \frac{- \begin{vmatrix} a_{n-1} & a_{n-5} \\ b_1 & b_3 \end{vmatrix}}{b_1} = \frac{b_1 a_{n-5} - b_3 a_{n-1}}{b_1} \quad [2.13]$$

The other rows follow in the same form. Before we advance to the use of this procedure, we shall establish an array for the following polynomial:

$$p(s) = 2s^4 + 3s^3 + 5s^2 + 2s + 6 \quad [2.14]$$

First, the initial part of the array is formed. The powers of s are written to the left, and the polynomial coefficients are alternated between the first and second rows, as shown. It is helpful to imagine the rows to continue to the right with entries of zeros.

$$\begin{array}{c} s^4 \\ s^3 \\ s^2 \\ s^1 \\ s^0 \end{array} \left| \begin{array}{ccc} 2 & 5 & 6 \\ 3 & 2 & \end{array} \right.$$

The array is completed by proceeding, row by row. Each element calculated is derived from four elements in the above two rows, two of them at the left column and two in the column to the right of the element being calculated.

For the example of Equation (2.14), the first element of the s^2 row is

$$b_1 = \frac{3(5) - 2(2)}{3} = 11/3$$

And the second term of the s^2 row is

$$b_2 = \frac{3(6) - 2(0)}{3} = 6$$

The first element of the s^1 row is

$$c_1 = \frac{(11/3)2 - 3(6)}{11/3} = -32/11$$

and the second element of the s^1 row is

$$c_2 = 0$$

The first element of the s^0 row is

$$d_1 = \frac{-(32/11)(6) - 0}{-32/11} = 6$$

Repeating pattern of calculations.

This is shown in the shaded areas as follows:

$$\begin{array}{c}
 s^4 \\
 s^3 \\
 s^2 \\
 s^1 \\
 s^0
 \end{array}
 \left| \begin{array}{c}
 \text{shaded} \\
 \text{shaded} \\
 \frac{11}{3} \\
 \text{shaded} \\
 \text{shaded}
 \end{array} \right.
 \begin{array}{c}
 6 \\
 \\
 \\
 \\
 \end{array}
 \quad
 \begin{array}{c}
 s^4 \\
 s^3 \\
 s^2 \\
 s^1 \\
 s^0
 \end{array}
 \left| \begin{array}{c}
 \text{shaded} \\
 5 \\
 \text{shaded} \\
 \frac{11}{3} \\
 6
 \end{array} \right.
 \quad
 \begin{array}{c}
 s^4 \\
 s^3 \\
 s^2 \\
 s^1 \\
 s^0
 \end{array}
 \left| \begin{array}{c}
 2 \quad 5 \quad 6 \\
 \text{shaded} \\
 \text{shaded} \\
 -\frac{32}{11} \\
 \text{shaded}
 \end{array} \right.$$

The completed Routh array is shown below. The number of RHP (right half-plane) roots of $p(s)$ is the number of algebraic sign changes in the elements of the left column of the array, proceeding from top to bottom. For this example, there are two sign changes in the left column, as indicated with the arrows; therefore $p(s)$ has two RHP roots.

$$\begin{array}{c}
 s^4 \\
 s^3 \\
 s^2 \\
 s^1 \\
 s^0
 \end{array}
 \left| \begin{array}{ccc}
 2 & 5 & 6 \\
 3 & 2 & \\
 \frac{11}{3} & 6 & \\
 -\frac{32}{11} & & \\
 6 & &
 \end{array} \right.$$

If $p(s)$ is the denominator polynomial of a system's transfer function, that system is unstable.

2.5.3 Significance of the Array Coefficients

The Routh array contains coefficients for polynomials resulting from a sequence of polynomial divisions. The first row is an abbreviated form of the even powers of s present in the $p(s)$ of Equation (2.8):

$$E(s) = 2s^4 + 5s^2 + 6$$

The second row of the array, in effect, contains coefficients of the odd powers of s from $p(s)$:

$$D(s) = 3s^3 + 2s$$

Suppose we divide the lower-order polynomial $D(s)$ into the higher-order polynomial $E(s)$:

$$\begin{array}{r}
 \frac{2}{3}s \\
 3s^3 + 2s \overline{) 2s^4 + 5s^2 + 6} \\
 \underline{2s^4 + \frac{4}{3}s^2} \\
 \frac{11}{3}s^2 + 6
 \end{array}$$

Notice that the remainder

$$R_1(s) = \frac{11}{3}s^2 + 6$$

defines the s^2 row of the Routh array. Now, if we divide $R_1(s)$ into $D(s)$

$$\begin{array}{r} \frac{11}{3}s^2 + 6s \overline{) 3s^3 + 2s} \\ \underline{3s^3 + \frac{54}{11}s} \\ -\frac{32}{11}s \end{array}$$

The remainder

$$R_2(s) = -\frac{32}{11}s$$

defines the s^1 row of the Routh array. Finally, the s^0 row of the Routh array is the remainder when $R_2(s)$ is divided into $R_1(s)$.

$$\begin{array}{r} -\frac{121}{96}s \overline{) \frac{11}{3}s^2 + 6} \\ \underline{\frac{11}{3}s^2} \\ 6 \end{array}$$

Notice that if a remainder is zero, an entire row of the Routh array is zero. If, for example, $R_2(s)$ had been zero, then $R_1(s)$ would have been a factor of $p(s)$. That fact will be exploited when we consider a procedure for handling a row of zeros.

As another example, a system with transfer function

$$T(s) = \frac{5s^2 - 7s + 2}{s^4 + 2s^3 + 3s^2 + 4s + 1}$$

has characteristic polynomial

$$s^4 + 2s^3 + 3s^2 + 4s + 1$$

which has the Routh array of Equation (2.15). There are no algebraic sign changes in the left column, so the polynomial has no RHP roots.

$$\begin{array}{c|c} s^4 & 1 \\ s^3 & 2 \\ s^2 & 3 \\ s^1 & 4 \\ s^0 & 1 \end{array}$$

[2.15]

Each array has properties that serve as a partial check on its correct completion. The number of nonzero row entries is normally reduced by one every two rows, with just one nonzero element in the s^1 row and in the s^0 row. And the last coefficient of the polynomial appears periodically as the last nonzero entry in every other row, as shown in the array of Equation (2.15).

2.5.4 Left-Column Zeros

Sometimes the polynomial coefficients are such that a zero occurs in the left column of the array, so that the array cannot be completed. The situation of having a zero

at the left of a row, although the entire row does not consist of zeros, is termed a *left-column zero*. For example, the polynomial

$$p(s) = 3s^4 + 6s^3 + 2s^2 + 4s + 5 \quad [2.16]$$

has an array that begins as follows:

$$\begin{array}{l|ll} s^4 & 3 & 2 & 5 \\ s^3 & 6 & 4 & \\ s^2 & 0 & 5 & \\ s^1 & & & \\ s^0 & & & \end{array}$$

The array cannot be completed in the usual way, because of the necessity to divide by zero. This zero in the first column can be eliminated by extending the array. The row where the first column 0 occurs is renamed the *A* row, and a new row, termed the *B* row, is formed from the *A* row. The *B* row is formed by sliding the *A* row to the left until the zeros disappear. The sign of the row is changed by $(-1)^n$ where n is the number of times this row is shifted to the left to eliminate the zeros. In the example, the *A* and *B* rows are as follows because $n = 1$,

Row manipulations solve the left-column zero problem.

$$\begin{array}{l|ll} A & 0 & 5 \\ B & -5 & 0 \end{array} \quad [2.17]$$

The new s^2 row is formed by adding the *A* and *B* rows together, with the result

$$s^2 \begin{array}{|l} -5 \quad 5 \end{array} \quad [2.18]$$

After the remaining terms are calculated as before, the entire array becomes

$$\begin{array}{l|lll} s^4 & 3 & 2 & 5 \\ s^3 & 6 & 4 & \\ s^2 & -5 & 5 & \\ s^1 & +10 & & \\ s^0 & 5 & & \end{array} \quad [2.19]$$

From the array of Equation (2.19), there are two sign changes— $+6$ to -5 and -5 to $+10$ —and hence there are two roots in the RHP.

Some years ago, another procedure was used when a left column zero was encountered, the so-called epsilon method. The new “row-shifting” method was found to work when the epsilon method would not work, so the epsilon method is not presented here. The article by Benidir and Picinbond cited at the end of this chapter describes both the epsilon method and the row-shifting method.

The epsilon method also solves the left-column zero problem.

Another example is presented, with the following polynomial:

$$p(s) = s^5 + s^4 + 2s^3 + 3s^2 + s + 4 \quad [2.20]$$

The array is formed in the normal fashion as follows:

$$\begin{array}{c|ccc} s^5 & 1 & 2 & 1 \\ s^4 & 1 & 3 & 4 \\ s^3 & -1 & -3 & \\ s^2 & 0 & 4 & \end{array} \quad [2.21]$$

The s^2 row now becomes the A row and the B row is formed by sliding the A row to the left one column and multiplying by -1 , with the result

$$\begin{array}{c|cc} A & 0 & 4 \\ B & -4 & 0 \end{array} \quad [2.22]$$

These are added to form the s^2 row and the array is completed in the usual fashion as follows:

$$\begin{array}{c|ccc} s^5 & +1 & 2 & 1 \\ s^4 & +1 & 3 & 4 \\ s^3 & -1 & -3 & \\ s^2 & -4 & 4 & \\ s^1 & -4 & 0 & \\ s^0 & +4 & & \end{array} \quad [2.23]$$

With two sign changes, from $+1$ to -1 and from -4 to $+4$, the polynomial of Equation (2.20) has two RHP roots.

2.5.5 Row of Zeros

An entire row of zeros occurs in the Routh array, when $p(s)$ contains an even polynomial as a factor, so, no remainder polynomial occurs in an odd-ordered row. For example, the test of the polynomial

$$p(s) = s^5 + 2s^4 + 8s^3 + 11s^2 + 16s + 12 \quad [2.24]$$

has a row of zeros at s^1 row:

$$\begin{array}{c|ccc} s^5 & 1 & 8 & 16 \\ s^4 & 2 & 11 & 12 \\ s^3 & \frac{5}{2} & 10 & \\ s^2 & 3 & 12 & \\ s^1 & 0 & 0 & \\ s^0 & & & \end{array} \quad [2.25]$$

Significance of a row of zeros.

Since the row of zeros indicates an even polynomial divisor of the original polynomial, the coefficients of that polynomial are those given in the row above the row of

zeros:

$$p_{\text{divisor}}(s) = 3s^2 + 12 = 3(s^2 + 4) = 3(s + 2j)(s - 2j)$$

The tested polynomial thus has two imaginary axis roots at $s = \pm j2$ and can be divided as follows:

$$\begin{array}{r}
 s^3 + 2s^2 + 4s + 3 \\
 s^2 + 4 \overline{) s^5 + 2s^4 + 8s^3 + 11s^2 + 16s + 12} \\
 \underline{s^5 + 4s^3} \\
 2s^4 + 4s^3 + 11s^2 + 16s + 12 \\
 \underline{2s^4 + 8s^2} \\
 4s^3 + 3s^2 + 16s + 12 \\
 \underline{4s^3 + 16s} \\
 3s^2 + 12 \\
 \underline{3s^2 + 12} \\
 0
 \end{array}$$

To complete the array, the row of zeros is replaced by the coefficients of the derivative of the divisor polynomial:

$$\frac{dp_{\text{divisor}}(s)}{ds} = 6s$$

s^5	1	8	16
s^4	2	11	12
s^3	$\frac{5}{2}$	10	
s^2	3	12	
s^1	6		
s^0	12		

We test the ($n = 2$)-order divisor by looking for any left-column sign changes below the dashed line. There are none, and there are two complex conjugate roots located at $s = \pm j2$ and no RHP or LHP roots. The remaining three roots are tested by using the three left-column signs, starting with the coefficient "1" and ending with the coefficient "3". Notice that the coefficient "3" is used twice: once to initiate a test of $E(s)$ and again to complete the test for the remaining roots. There are no sign changes, so the remaining roots must be in the LHP.

Even and odd polynomials have root locations that are symmetric about the imaginary axis. An odd polynomial always has s as a factor. Whether or not s is a factor is evident from a glance at the polynomial, and the remainder is an even polynomial. For any even polynomial, replacing s with $(-s)$ leaves the polynomial unchanged. Thus, besides occurring in conjugate pairs, the roots of an even polynomial are also symmetrical about the imaginary axis. There are three basic types of factor possible in an even polynomial. One type is

$$(s + ja)(s - ja) = (s^2 + a^2)$$

Test below the dashed line separately from the top of the array.

which consists of complex conjugate roots on the imaginary axis, as indicated in Figure 2.11(a). Another is

$$(s + a)(s - a) = (s^2 - a^2)$$

which are symmetrical roots on the real axis, one in the LHP and one in the RHP, as in Figure 2.11(b). the third type of factor, Figure 2.11(c), involves complex roots in quadrature, one pair in the LHP and one pair in the RHP:

$$\begin{aligned} & (s + a + jb)(s + a - jb)(s - a + jb)(s - a - jb) \\ &= s^4 + 2(b^2 - a^2)s^2 + (a^2 + b^2)^2 \end{aligned}$$

The additional symmetry of even polynomial roots about the imaginary axis allows determination of the number of imaginary axis roots. Each RHP root of an even polynomial must be matched by just one corresponding LHP root. Thus if an even polynomial is of sixth order and is known to have just one RHP root, it has just one LHP root, and the remaining four roots must be on the imaginary axis. If an eighth-order polynomial has three RHP roots, it must have three LHP roots and two imaginary axis roots. For example, for the polynomial

$$s^6 + s^5 + 5s^4 + s^3 + 2s^2 - 2s - 8 \quad [2.26]$$

the coefficient tests indicate the presence of at least one RHP root. The Routh-Hurwitz test begins as follows:

$$\begin{array}{l|llll} s^6 & 1 & 5 & 2 & -8 \\ s^5 & 1 & 1 & -2 & \\ s^4 & 4 & 4 & -8 & \\ s^3 & 0 & 0 & & \end{array}$$

The row of zeros means there is an even divisor.

So

$$4s^4 + 4s^2 - 8$$

is a divisor of the original polynomial. Replacing the row of zeros by the coefficients of the derivative of the divisor polynomial, we write

$$\frac{d}{ds} (4s^4 + 4s^2 - 8) = 16s^3 + 8s$$

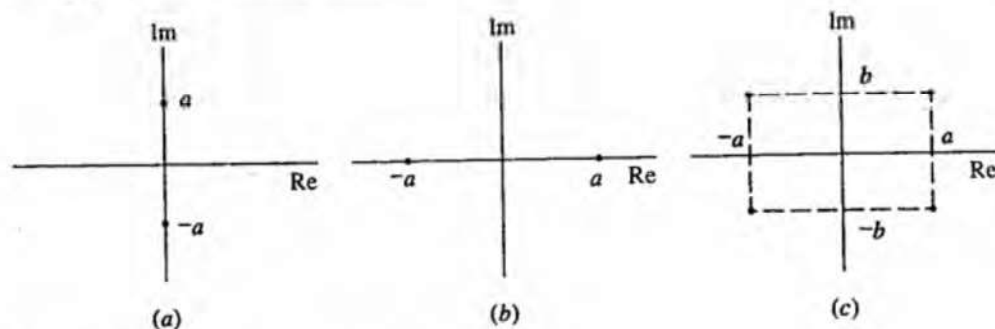


Figure 2.11 Even polynomial root locations.

The completed array is as follows:

$$\begin{array}{c|cccc}
 s^6 & 1 & 5 & 2 & -8 \\
 s^5 & 1 & 1 & -2 & \\
 \hline
 s^4 & 4 & 4 & -8 & \\
 s^3 & 16 & 8 & & \\
 s^2 & 2 & -8 & & \\
 s^1 & 72 & & & \\
 s^0 & -8 & & &
 \end{array} \quad [2.27]$$

The test of the even polynomial divisor is that portion of the array below the dashed line.

Below the dashed line for the ($n = 4$)-order even polynomial, there is one root in the RHP and one root in the LHP. There are two roots on the imaginary axis. The remaining two roots are tested by using the top two sign changes determined by the top three rows (a maximum of two possible sign changes). Since there are no such sign changes, the remaining two roots are in the LHP, and we have the following numbers of the roots of various types:

Test below the dashed line separated from the top of the array.

$$\text{RHP} = 1$$

$$\text{LHP} = 3$$

$$\text{IA} = 2$$

Another array with an all-zero row is the following:

$$\begin{array}{c|ccc}
 s^4 & 1 & 9 & 20 \\
 s^3 & 6 & 24 & \\
 \hline
 s^2 & 5 & 20 & \\
 s^1 & 0 & 0 & \\
 s^0 & & &
 \end{array}$$

The s^1 is an all-zero row, so the system has a divisor found from the s^2 row as follows:

$$5s^2 + 20 = 0 = 5(s^2 + 4) \quad [2.28]$$

From Equation (2.28), the roots on the j axis are at $\pm j2$. Replacing the row of zeros by the coefficients of the derivative of the divisor polynomial, we obtain $d/ds(5s^2 + 20) = 10s$. The array is continued with the result

$$\begin{array}{c|ccc}
 s^4 & 1 & 9 & 20 \\
 s^3 & 6 & 24 & \\
 \hline
 s^2 & 5 & 20 & \\
 s^1 & 10 & & \\
 s^0 & 20 & &
 \end{array}$$

For the ($n = 2$) order even divisor, there are no sign changes below the dashed line, so no RHP roots. Two roots are on the imaginary axis located at $\pm j2$, hence there are

no LHP roots. Since there are no sign changes above the dashed line, there must be two roots in the LHP. The roots are as follows:

$$\text{RHP} = 0$$

$$\text{LHP} = 2$$

$$\text{IA} = 2$$

The polynomial

$$s^5 + s^4 + 6s^3 + 6s^2 + 25s + 25$$

has an all-zero row at the s^3 row. The system is continued by differentiating the s^4 row with the result:

s^5	1	6	25
s^4	1	6	25
s^3	4	12	
s^2	3	25	
s^1	64		
s^0	3		
		25	

the fourth-order even polynomial divisor has two RHP roots, and so its remaining two roots must be in the LHP. There can thus be no imaginary axis roots. The entire polynomial has two RHP roots, and the remaining three roots must be in the LHP.

2.5.6 Eliminating a Possible Odd Divisor

Could an odd divisor appear in a Routh array? That is, could a row of zeros occur in an even-ordered row, causing an odd-ordered row above it to contain a divisor of $p(s)$? If there is an odd divisor of $p(s)$, then s must be divisible into $p(s)$ so that $p(s)$ would lack an a_0 coefficient.

When the a_0 coefficient is zero, we need only factor out an appropriate number of s values and test what remains.

Suppose we want to test

$$\begin{aligned} p(s) &= s^5 + 3s^4 + 2s^3 + 4s \\ &= sp_1(s) = s[s^4 + 3s^3 + 2s^2 + 4] \end{aligned}$$

Instead, test

$$p_1(s) = [s^4 + 3s^3 + 2s^2 + 4]$$

and remember that one additional root of $p(s)$ exists at $s = 0$. Once it has been agreed to remove any s factors, a row of zeros can happen only in an odd-ordered row of a Routh array, revealing an even divisor of $p(s)$ above that row of zeros.

2.5.7 Multiple Roots

Assuming that we have acted to exclude odd divisors, if multiple roots are present, the even divisor must, itself, have an even divisor, meaning that the Routh array must reveal a row of zeros twice. When there is one row of zeros, at best the system is marginally stable if some roots are in the IA and the rest are in the LHP. When multiple roots are present, either RHP roots are multiple or IA roots are multiple. In any case, if a row of zeros occurs more than once, the system will be unstable.

Consider the polynomial

$$p(s) = s^6 + 5s^5 + 14s^4 + 40s^3 + 64s^2 + 80s + 96$$

We begin by forming the Routh array and proceed with row evaluations. Notice that all polynomial coefficients are positive; but no stability implications should be made.

$$\begin{array}{c|cccc} s^6 & 1 & 14 & 64 & 96 \\ s^5 & 5 & 40 & 80 & \\ s^4 & 6 & 48 & 96 & \\ s^3 & 0 & 0 & & \end{array}$$

Since the s^3 row contains all zeros for the remainder polynomial, the system has an even divisor.

$$E_1(s) = 6s^4 + 48s^2 + 96 = 6[s^4 + 8s^2 + 16]$$

To complete part of the rest of the array, replace the zeros with the coefficients of

$$\frac{dE_1(s)}{ds} = 24s^3 + 96$$

$$\begin{array}{c|cccc} s^6 & 1 & 14 & 64 & 96 \\ s^5 & 5 & 40 & 80 & \\ s^4 & 6 & 48 & 96 & \\ s^3 & 24 & 96 & & \\ s^2 & 24 & 96 & & \\ s^1 & 0 & & & \end{array}$$

Now, $E_1(s)$ must have a divisor given by

$$E_2(s) = 24s^2 + 96 = 24[s^2 + 4]$$

Since

$$\frac{dE_2(s)}{ds} = 48s$$